

7. GOVERNANCE, LINEAGE & RELEASE GATE

STPA Hazard Mitigation · Bumpless Human Joystick Override · Defensible Release Case (LOOP-01 → REL-01)

1. SENSING

Photons / Voltages to DMA
MIPI CSI-2 · I2C · SPI Bus Priority

Raw Frames

2. PERCEPTION & ENCODERS

Raw Tensors to Spatial Tokens
ViT Encoders · DINOv2 · 3D Affordances

3D Tokens

3. WORLD MODELS & STATE

Latent Belief & SE(3) Frame Trees
JEPA / RSSM Dynamics · TTL Validity Leases

State Vector

SYSTEM 1.5 · LINUX MPU / NPU

5. TRAJECTORY DECODERS

Diffusion Policies · ACT Action Chunking (H -steps) · C^2 Jerk Continuity

Intent Goal

SYSTEM 2 · LINUX MPU

4. SEMANTIC DELIBERATION

Vision-Language Foundation Models · Open-Vocabulary Goals · Expiring Leases (t_{expire})

Expiring Proposal p_t

THE PROPOSAL-PERMISSION PRIVILEGE BOUNDARY (NO DIRECT MOTOR ACCESS)

SYSTEM 1 · REAL-TIME MCU (BARE-METAL / FreeRTOS)

6. REAL-TIME REFLEX & SAFETY ENFORCEMENT

1 kHz Reflex Timing Loop · Control Barrier Functions ($h(x) \geq 0$) · Dynamic Stopping Distance $d_{\text{stop}}(v_t) \leq d_{\text{clearance}}$ · Hardware Veto (u_t)

Permitted Action $u_t = \text{permit}(p_t)$

THE PHYSICAL WORLD ($W_t \rightarrow W_{t+1}$)

Kinetic Momentum ($p = mv$) · Joule Heat Dissipation (I^2R) · Matter Mutation · Friction (μ) · Collision Dynamics

Endogenous Sensory Shift ($A_t \rightarrow W_{t+1} \rightarrow O_{t+1}$)